

HANDOUTS

115.109

Introduction to Business Analytics

Topic 9: Relationships

Dr Andrew Murphy

(Please contact me via [Communication Tools in Stream](#))

Week (starting)	Business Problem	Data & Analysis	Chapter (Black et al)	Assess ments
1 (13/7)	Understanding Data: discounts & margin	Calculating data in Excel; GST, discounts and Margins	1	Lab1
2 (20/7)	Operational Monitoring: Dashboards	Tabulating & Graphing	2	Lab2
3 (27/7)	Industry trends	Statistics NZ: retail survey Descriptive stats: averages	3	Lab3
4 (3/8)	Customer trends	Statistics NZ: census Descriptive stats: variation	3	Lab4
5 (10/8)	Transaction data: product sales	Transaction data sources. Sampling	7	Lab5
6 (17/8)	Loyalty cards	Loyalty card schemes. Probability; proportions, χ^2	4, 12	Lab6 Test 1
Mid	Semester	Study	Period	
7 (7/9)	Segmentation	Statistics NZ: census. Normal distribution; Means	6, 8	Lab7
8 (14/9)	Determining difference (1)	Means; ANOVA	9, 10, 11	Lab8
9 (21/9)	Associating and Predicting	Explaining relationships: Correlation & regression	13	Lab9
10 (28/9)	Determining Difference (2)	Explaining relationships (continued)	14	-
11 (5/10)	Trends over time	Time series, Indices Unstructured data: social media	15	Lab11 Test 2
12 (12/10)	Corporate reporting	Finance Dashboards reporting		Report

Assessments Schedule

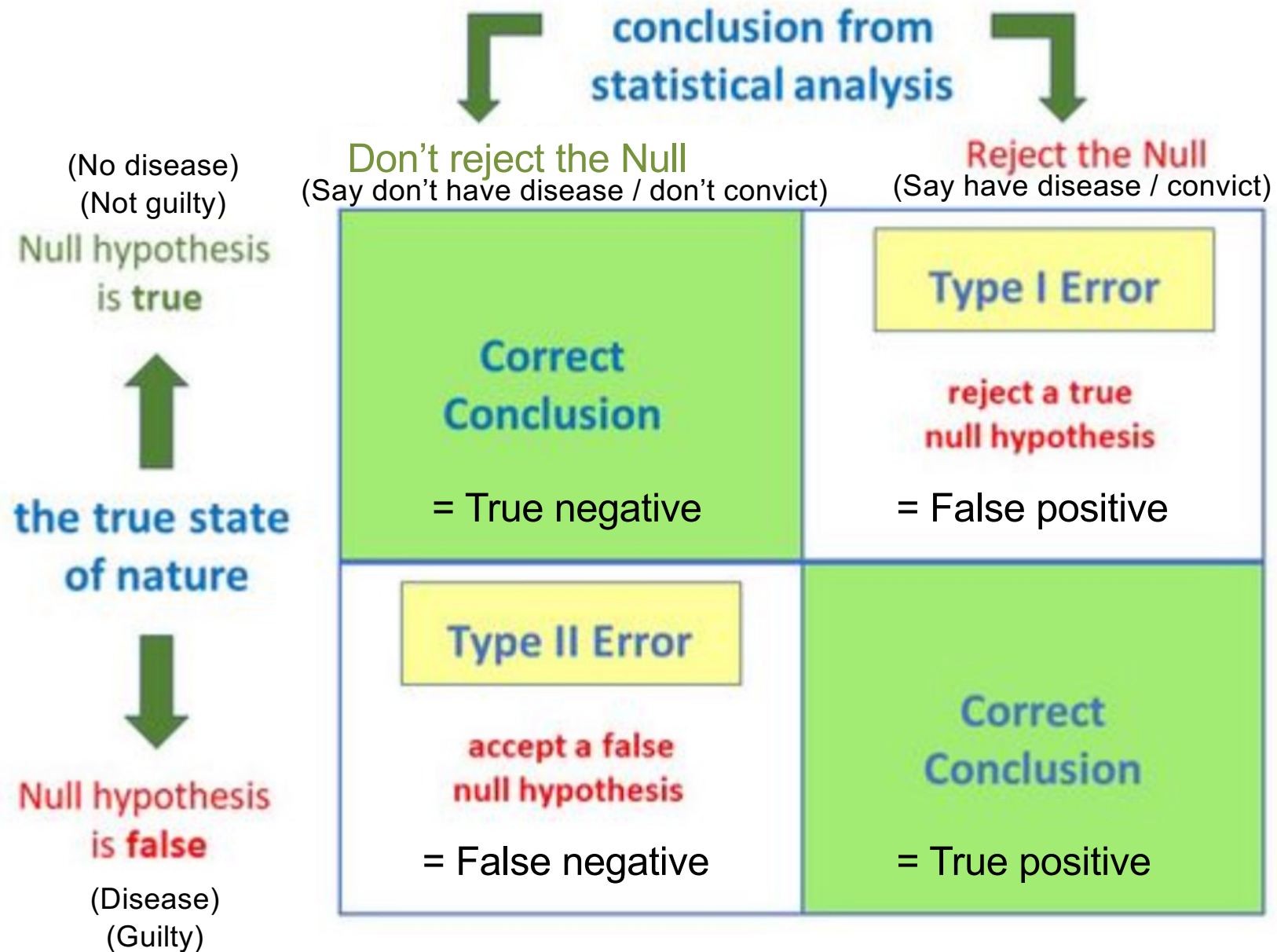
2026	Assessment	Lab 1&2	Lab 3&4	Lab 5&6	Test 1	Lab 7&8	Lab 9,11	Test 2	Report
Week	13-Jul	O: 13.07							
commencing	20-Jul		O: 20.07						
Monday:	27-Jul								
	03-Aug	D: 03.08							
	10-Aug			O: 10.08					
	17-Aug		D: 17.08		O: 14.08 C: 21.08				O 17.08
	<i>Mid-Semester Study Period</i>			D:31.08					
	07-Sep					O: 07.09			
	14-Sep								
	21-Sep						O: 21.09		
	28-Sep								
	05-Oct					D: 05.10		O: 02.10 C: 09.10	
	12-Oct						D: 19.10		D: 19.10
O=opens (available from); D=due (without late penalty); C=closes (not available after)									

Type I and Type II errors

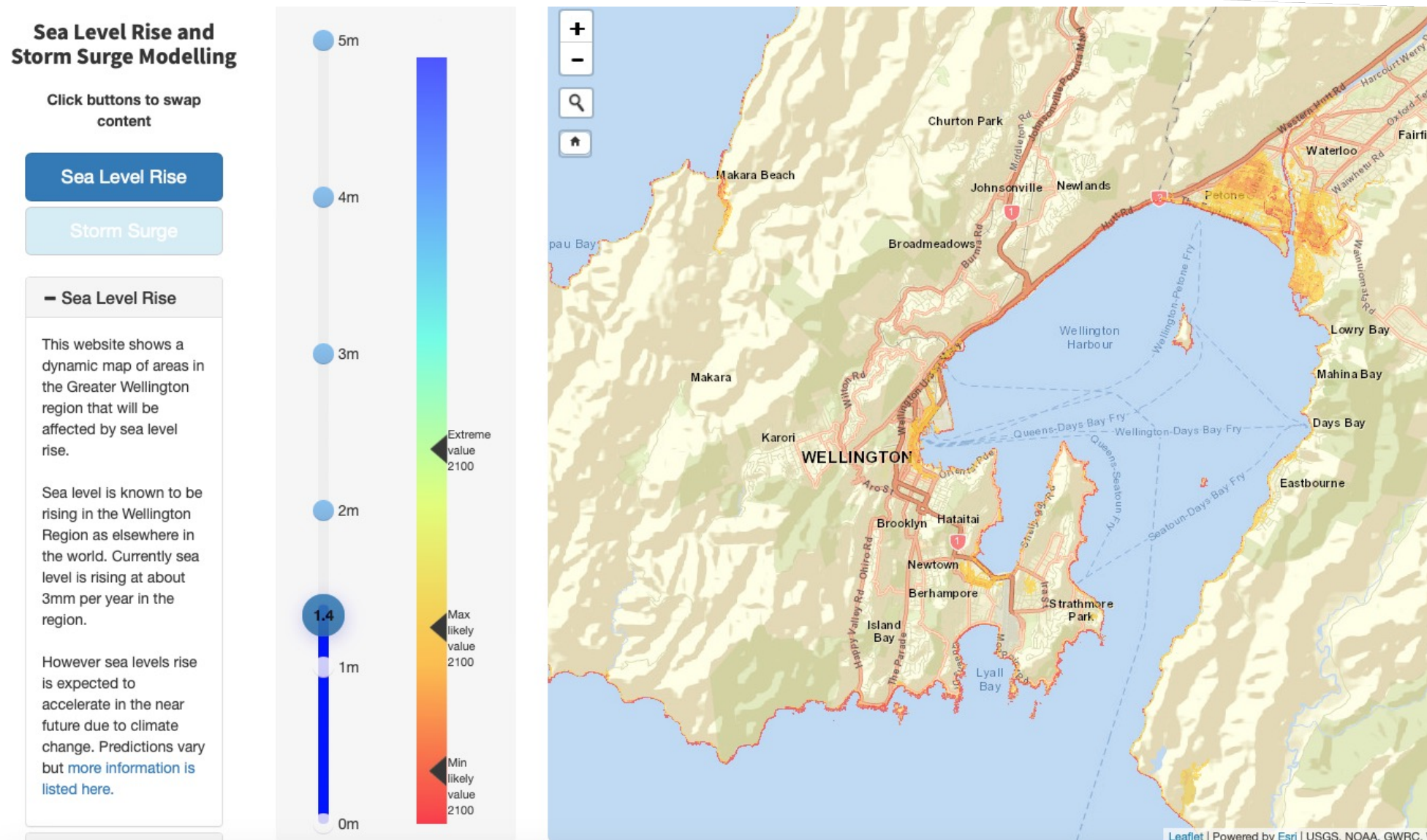
- Since hypothesis testing is based on probability theory, the researcher cannot be completely certain and runs the risk of committing one of two types of errors:

Type I error	Type II error
• An error caused by rejecting the null hypothesis when it is true • Probability of alpha (α)	• An error caused by failing to reject the null hypothesis when the alternative hypothesis is true • Probability of beta (β)

Type I and Type II errors



Modelling: predicting the future



<https://mapping1.gw.govt.nz/GW/SLR/>

Correlation

- The consideration of whether there is any relationship or association between two variables is called *correlation analysis*
- The *correlation coefficient* is the index which defines the strength of association between two variables
- There are many instances where you may want to test whether there is a relationship between two variables
 - *Examples:*
 - Time spent running on a treadmill and calories expended
 - Height and shoe size
 - the age of a machine and how often it breaks down
 - the level of education obtained by an individual and his or her income in later working life

Correlation

- If there is a relationship between any two variables, it may be possible to predict the value of one of the variables from the value of the other
- *Dependent and independent variables*
 - To establish whether there is a relationship between two variables, an appropriate random sample must be taken and measurement recorded for each of the two variables (*bi-variate*)
 - To predict the value of one variable from the value of the other (if a relationship exists), there is a basic rule
 - The *dependent variable* is the one whose value is to be predicted. It is usually denoted by the letter y
 - The *independent variable* is the one whose value is used to make the prediction. It is usually denoted by the letter x

Scatter Diagrams (X-Y Plots)

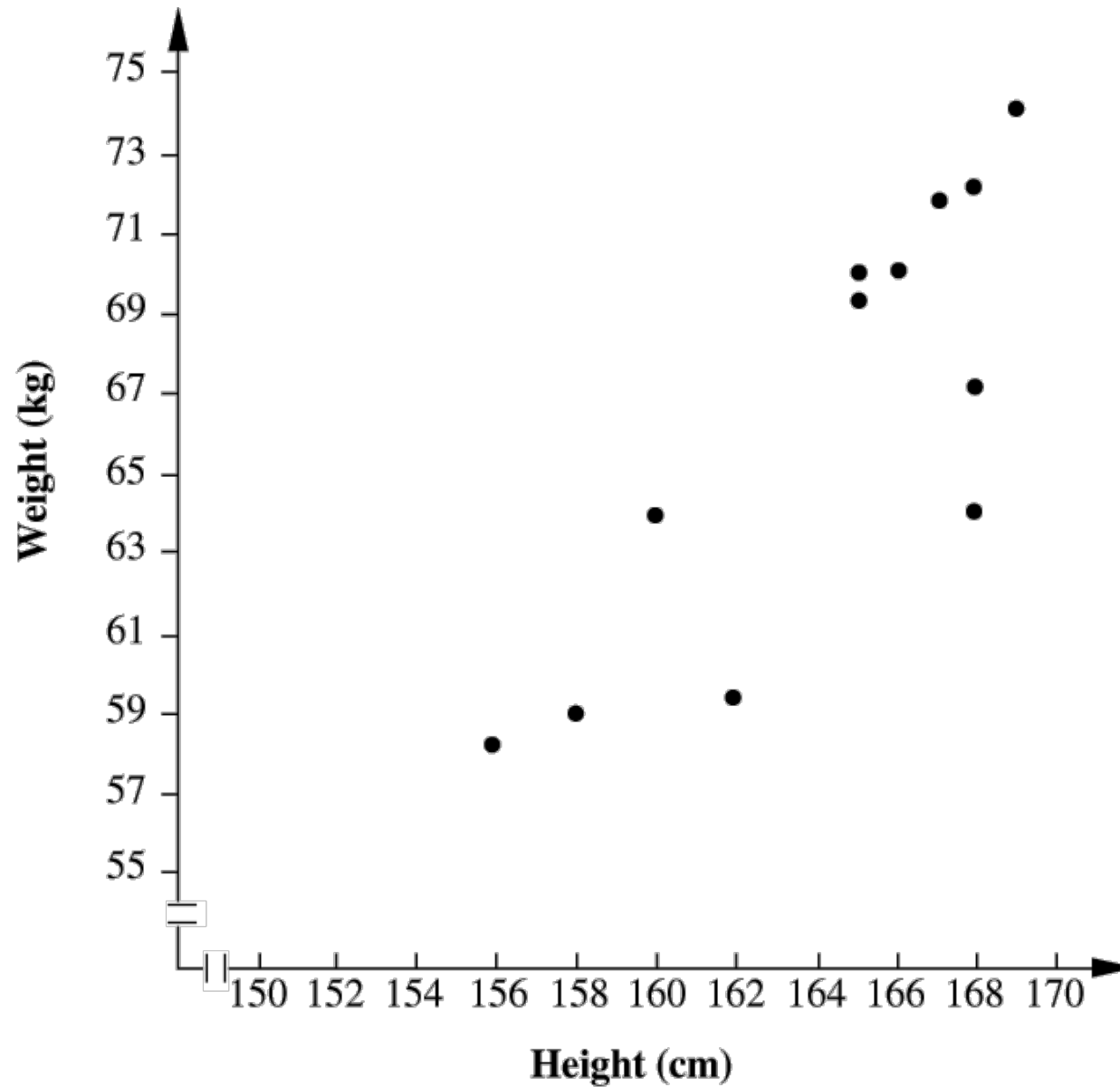
- A *scatter diagram* or *scatterplot* is a display in which ordered pairs of measurements are plotted on a coordinate axes system
 - The independent variable (x) is represented on the horizontal axis
 - The dependent variable (y) is represented on the vertical axis
- The points representing the data are usually plotted either by dots or crosses
 - Scatterplots can help get a ‘feel’ for what the actual relationship between the variables may be, even before any calculations are undertaken

Height vs Weight

- An insurance company manager is concerned about the health of young adult males, since the company is prepared to give a reduced premium rate to those who have a certain level of fitness. In particular, she would like to investigate how their height is related to their weight, with a view to possibly using these measurements as a fitness criterion
- To this end, she selects a random sample of 12 young adult males and measures both their height (in cm) and weight (in kg):

Number	1	2	3	4	5	6	7	8	9	10	11	12
Height (cm)	167	168	165	165	160	156	169	166	162	158	168	168
Weight (kg)	71.8	72.0	69.3	70.0	64.2	58.1	74.0	70.0	59.3	59.0	67.1	64.0

Scatterplot



Pearson product-moment correlation coefficient

- A numerical measure of the degree of association between two variables is given by the *product-moment correlation coefficient*
- This index provides a quantitative measure of the extent to which the two variables are associated
- The value of the correlation coefficient is calculated from the bivariate data by means of a formula that involves the values of the data points
- The value of the correlation coefficient calculated from a *sample* is denoted by the letter r

Pearson product-moment correlation coefficient

- Formulas, formulas

$$r = \frac{S_{xy}}{\sqrt{(S_{xx})(S_{yy})}}$$

or

$$r = \frac{\sum xy - n\bar{x}\bar{y}}{(n-1)S_x S_y}$$

$$S_{xx} = \sum (x - \bar{x})^2$$

$$S_{xx} = (n-1)S_x^2$$

or

$$S_{yy} = \sum (y - \bar{y})^2$$

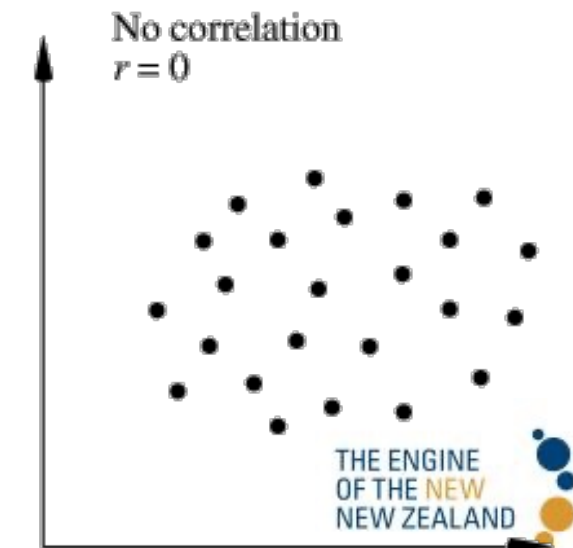
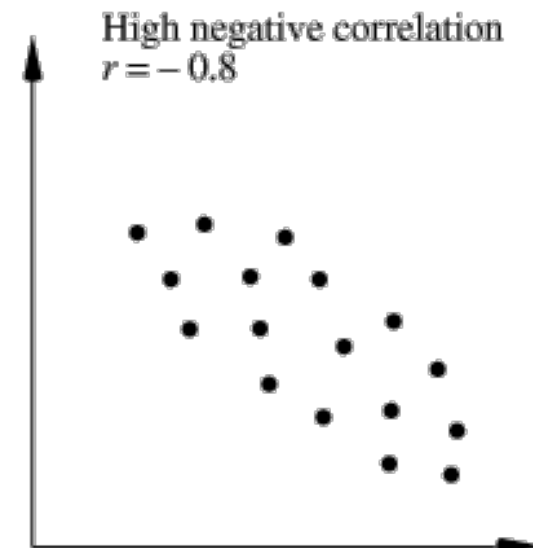
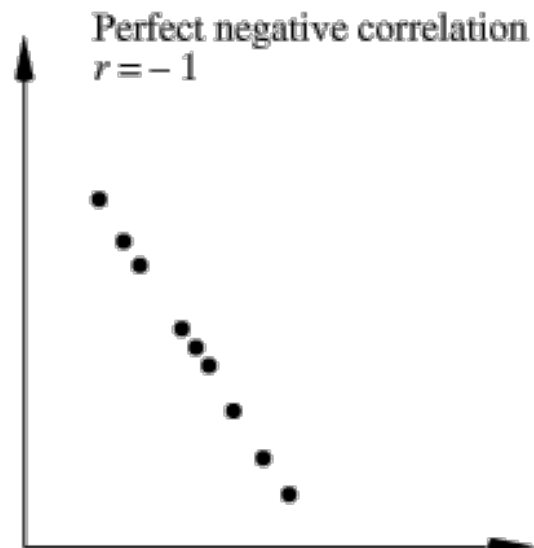
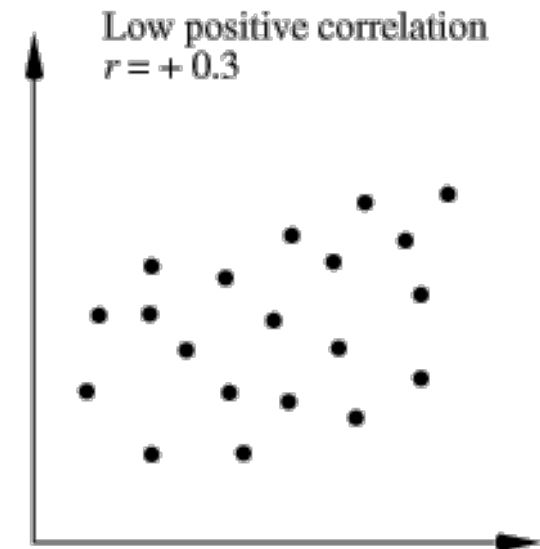
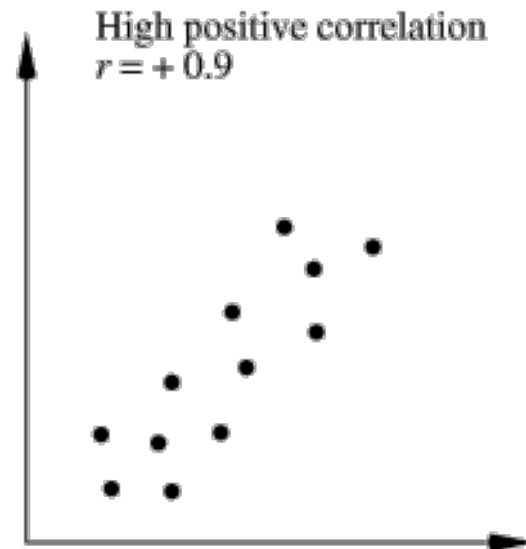
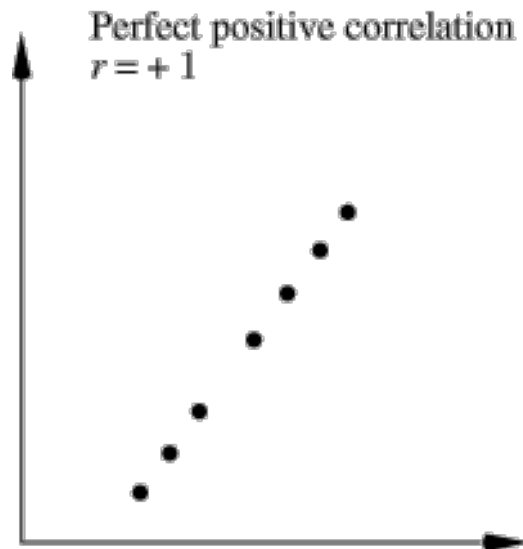
$$S_{yy} = (n-1)S_y^2$$

$$S_{xy} = \sum (x - \bar{x})(y - \bar{y})$$

Pearson product-moment correlation coefficient

- The value of r must *always* lie between -1 and $+1$ (both inclusive)
- If $r = +1$, the two variables have *perfect positive* correlation. This means that on a scatter diagram, the points all lie on a straight line that has a *positive* slope
- If $r = -1$, the two variables have *perfect negative* correlation. This means that on a scatter diagram, the points all lie on a straight line that has a *negative* slope
- If the two variables are *positively* correlated, but not perfectly so, the coefficient lies between 0 and 1
- If the two variables are negatively correlated, but not perfectly so, the coefficient lies between -1 and 0
- If the two variables have no overall upward or downward trend whatsoever, the coefficient is 0

Correlation



Pearson Correlation Coefficient

- *Calculation of r*

Example: weight by height

In this case, we denote *height* by x and *weight* by y

$$n = 12$$

$$\bar{x} = 164.3$$

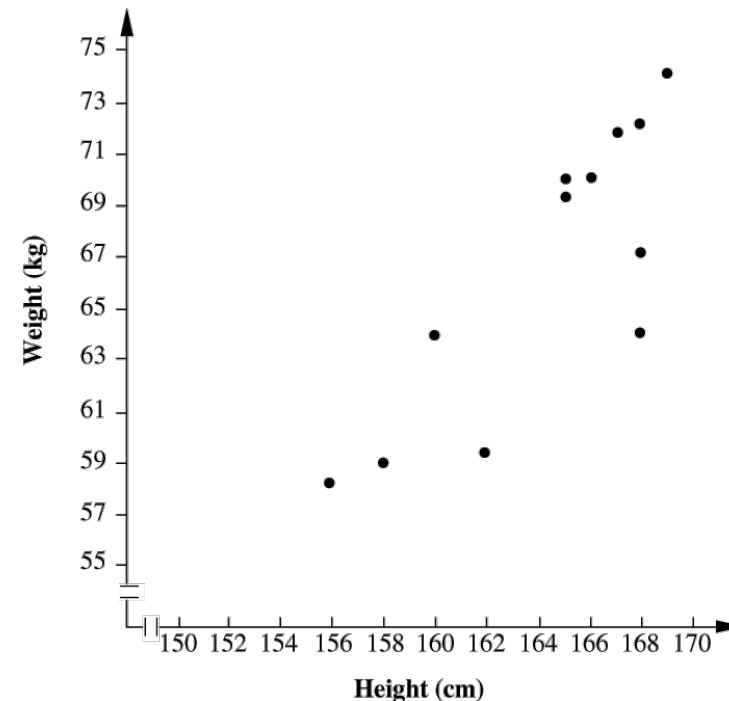
$$\bar{y} = 66.6$$

$$S_{xx} = \sum (x - \bar{x})^2 = 206.68$$

$$S_{yy} = \sum (y - \bar{y})^2 = 337.44$$

$$S_{xy} = \sum (x - \bar{x})(y - \bar{y}) = 216.62$$

$$r = \frac{216.62}{\sqrt{(206.68)(337.44)}} \\ = 0.82$$



Spurious correlation

- If two variables are significantly correlated, this does *not* imply that one must be the *cause* of the other
- The correlation coefficient is subject to variations in sampling
- Correlation between two variables that is really induced by other external variables is referred to as *spurious correlation* or *false correlation*
- Everyone who has ever eaten a carrot has either died, or will die in the future. Therefore carrots cause death.

Unusual claimed correlations

- There is no shortage of research in which some very odd relations are claimed to be found
 1. Attractiveness of a couple and the sex of their first child (physically attractive couples are 36% more likely than an unattractive couple to produce a girl as their first child)
 2. Height of children and mental development (short children perform more poorly on intelligence tests than tall ones)
 3. Blood flow to the heart and type of movie watched (watching comedy boosts blood flow; sad or distressing movies lower it)

Understanding Causation

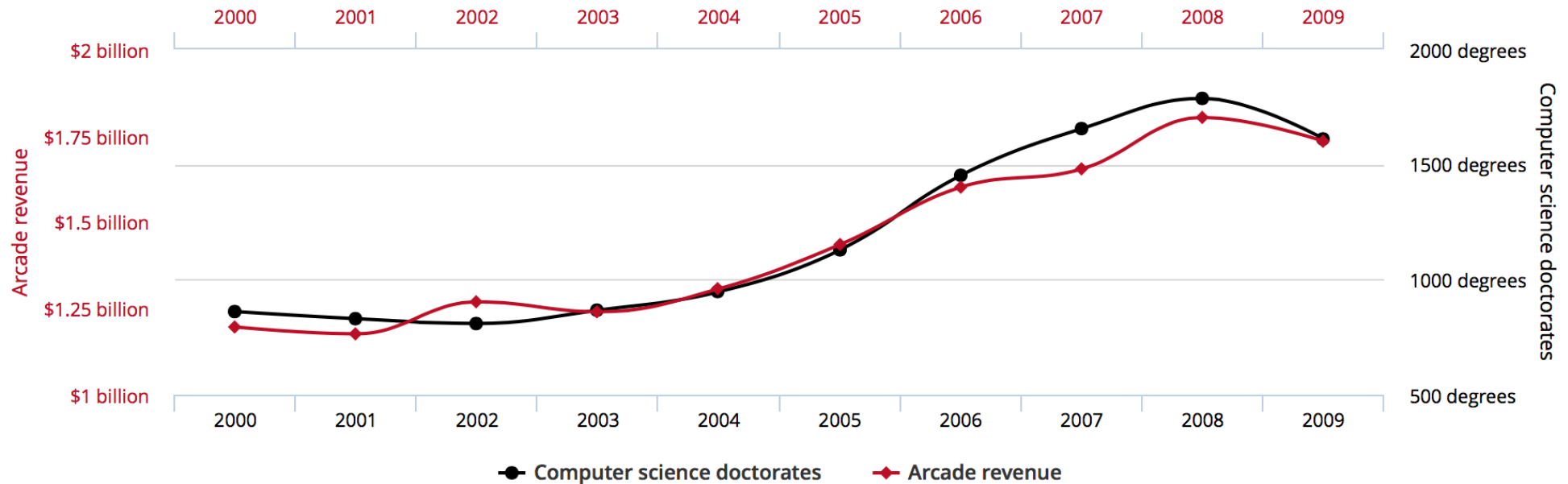
<https://www.youtube.com/watch?v=VMUQSMFGBDo>

Spurious Correlations

<http://tylervigen.com/spurious-correlations>

Total revenue generated by arcades correlates with Computer science doctorates awarded in the US

Correlation: 98.51% ($r=0.985065$)



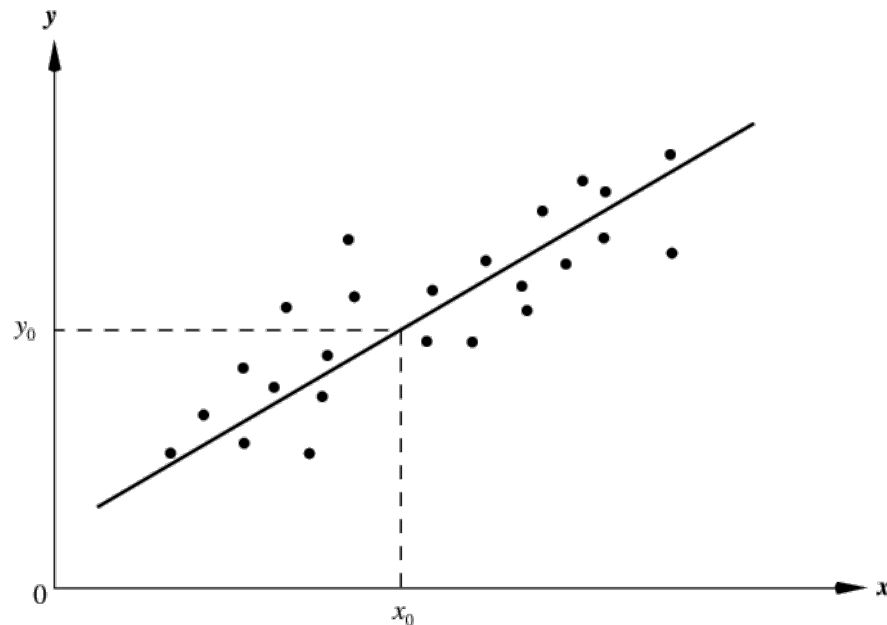
tylervigen.com

From Correlation to Regression

- If there is a significant correlation between two variables, it may be possible to predict the value of one of the variables (y) from knowledge of the value of the other (x)
 - This requires additional information: **causality**
- A predictive equation of this kind is called a *regression equation* or *regression line*
- The simplest is called *simple linear* regression: one 'x' variable is used to predict values of 'y'
 - Also have *multiple linear* regression (many 'x' variables) and *non-linear* regression

Simple Linear Regression

- *Linear regression* is the case where the relationship between the variables is represented by a straight line



- The dots are the actual data points graphed as a scatter diagram
- The straight line has been chosen to pass through the points as centrally as practicable
- AKA “least squares” regression

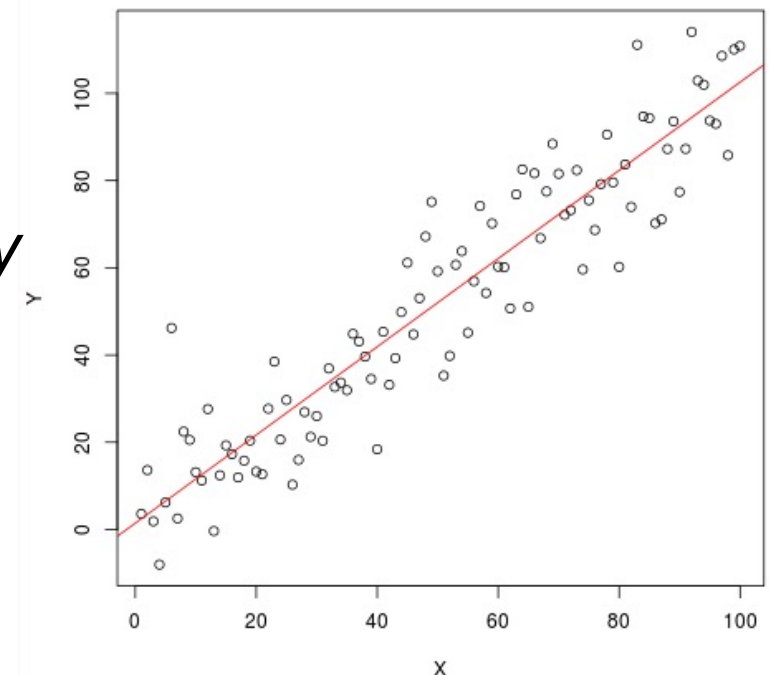
Regression: slope and intercept

- We are trying to find the line that will predict the value of the y -variable from a knowledge of the x -variable
- This is known as the *least-squares regression line* of y on x
- Has the form: $\hat{y} = \alpha + \beta x$ where:

α (alpha) is called the 'constant'

β (beta) is the 'parameter'

$\hat{y}$ (y-hat) is the predicted value of y



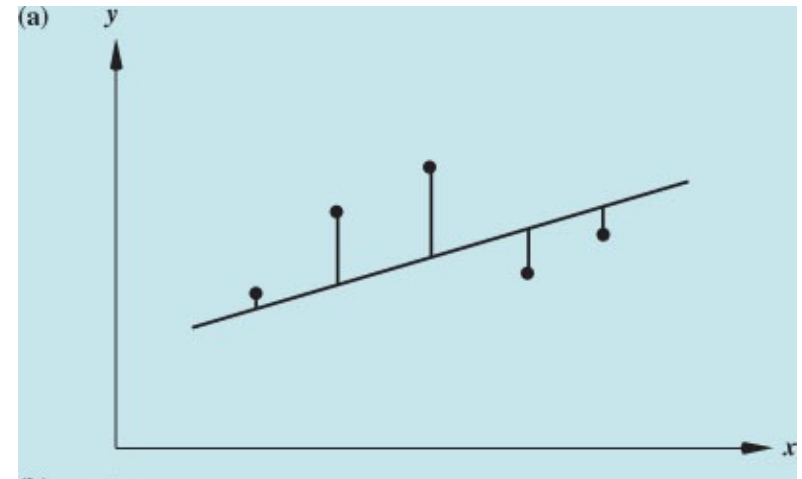
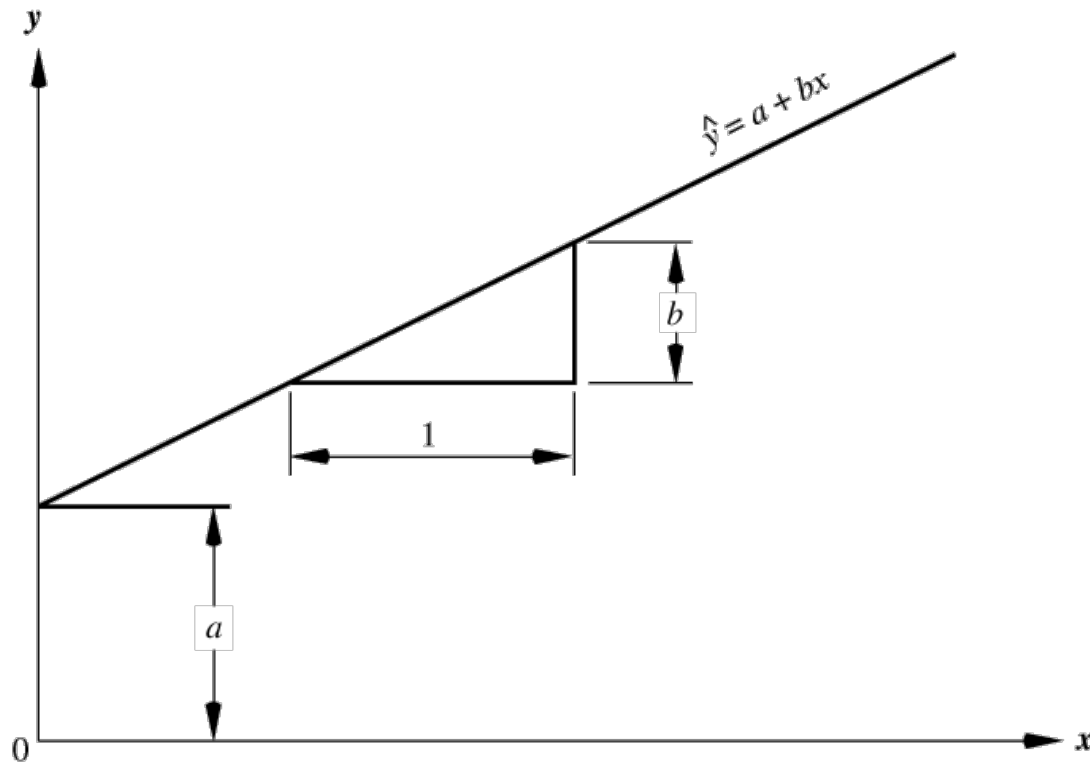
Regression: slope and intercept

- We usually never know the true values of α and β , since we cannot measure the entire population
- Instead, we use sample values (estimates) for x and y
- The least squares regression line of y on x for our sample may be written as:

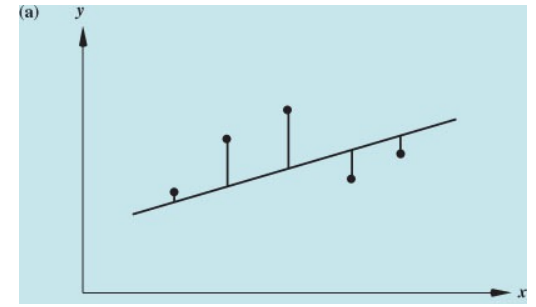
$$\hat{y} = a + bx$$

- The constant ' a ' is the value of y where the line intersects the y -axis, that is where $x = 0$
- The parameter value ' b ' denotes the *slope* of the line, representing the increase in the value of y when the value of x is increased by 1

Regression: slope and intercept



Least-squares regression



- The *vertical* distance from each point to the line represents *errors* that the line would make in predicting the value of y for the x -value of each data point
- The aim is to construct a line that will minimise these errors for the particular set of data points
- These errors (*residuals*) are the differences between the *actual* y -values and the *predicted* y -values $(y - \hat{y})$
(similar in principle to observed – expected)
- We want to find the straight line that will minimise the value of the expression $\sum (y - \hat{y})^2$, which is also called the *sum of the squares of the residuals*

Least-squares regression

- It can be shown that the values of b that minimise this sum of the squares of the residuals are given by:

$$b = \frac{S_{xy}}{S_{xx}} \text{ where:}$$

$$S_{xy} = \sum (x - \bar{x})(y - \bar{y})$$

$$S_{xx} = \sum (x - \bar{x})^2$$

$\bar{x}$ = mean of x – values

$\bar{y}$ = mean of y – values

Least-squares regression

- Note that the values of S_{xy} and S_{xx} are the same as those found when calculating the value of the product-moment correlation coefficient, r
- Once the value of b is determined, the value of a can be found from:

$$a = \bar{y} - b\bar{x}$$

continued

Example: Least-squares

A car manufacturer believes that the quarterly spare-parts repair bill for a popular model of car is strongly influenced by the age of the car. A random sample of 8 cars of varying ages is chosen and their repair bills for the previous 12 months are recorded. The results are:

Age (years) = X	Quarterly repair bill (\$) = Y	$x - \bar{x}$	$y - \bar{y}$
1	70	-5	-210
3	140	-3	-140
5	230	-1	-50
8	350	2	70
7	300	1	20
12	570	6	290
8	380	2	100
4	200	-2	-80

$$\bar{x} = \frac{\sum x}{n} = \frac{48}{8} = 6$$

$$\bar{y} = \frac{\sum y}{n} = \frac{2240}{8} = 280$$

Example: least-squares

$$\bar{x} = \frac{\sum x}{n} = \frac{48}{8} = 6$$

$$\bar{y} = \frac{\sum y}{n} = \frac{2240}{8} = 280$$

$$S_{xy} = \sum (x - \bar{x})(y - \bar{y}) = 3780$$

$$S_{xx} = \sum (x - \bar{x})^2 = 84$$

$$b = \frac{S_{xy}}{S_{xx}} = \frac{3780}{84} = 45$$

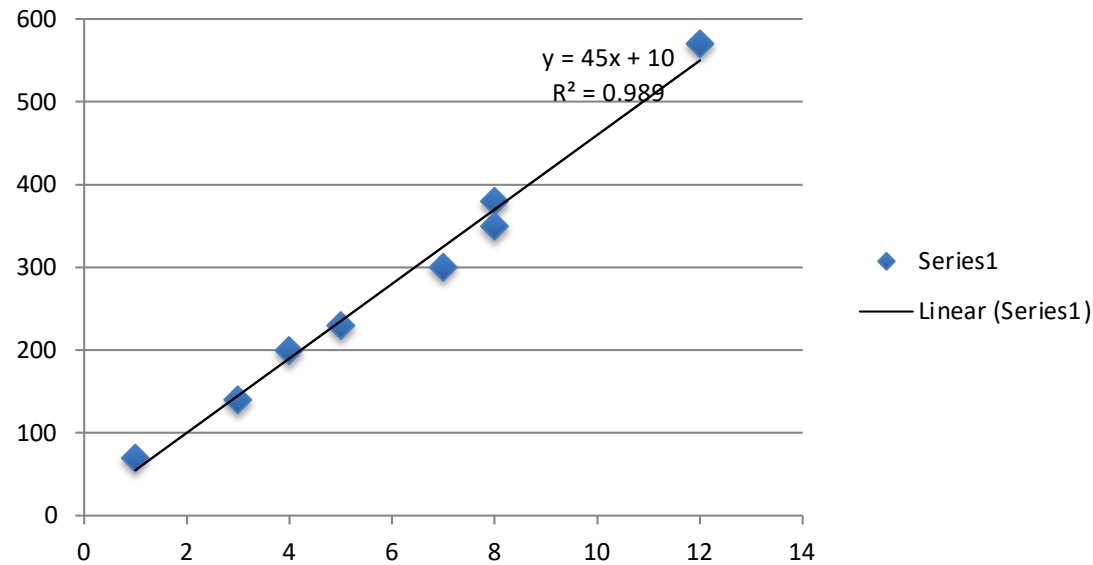
$$\begin{aligned} a &= \bar{y} - b\bar{x} \\ &= 280 - (45)(6) \\ &= 10 \end{aligned}$$

$$\hat{y} = 10 + 45x$$

Quarterly repair bill (\$) = 10 + 45 (age of car yrs)

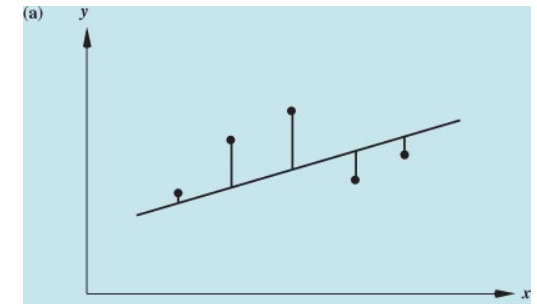
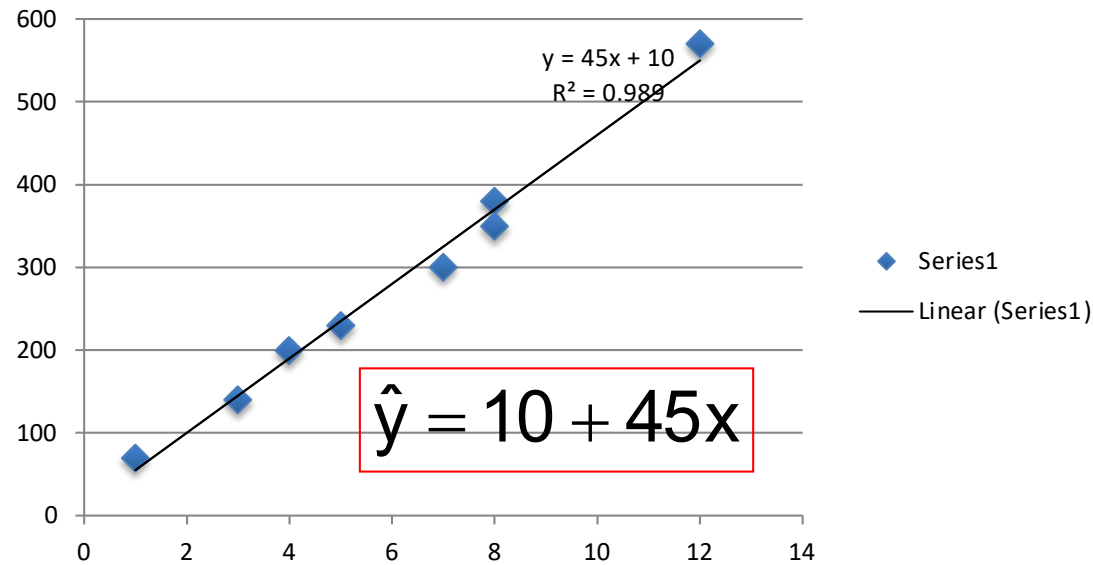
	X	Y	X - $\bar{x}$	Y - $\bar{y}$	(X - $\bar{x}$) (Y - $\bar{y}$)	(X - $\bar{x}$) (X - $\bar{x}$)
	1	70	-5	-210	1050	25
	3	140	-3	-140	420	9
	5	230	-1	-50	50	1
	8	350	2	70	140	4
	7	300	1	20	20	1
	12	570	6	290	1740	36
	8	380	2	100	200	4
	4	200	-2	-80	160	4
Mean:	6	280		Sum:	3780	84

Example: least-squares



	X	Y	$X - \bar{X}$	$Y - \bar{Y}$	$(X - \bar{X})(Y - \bar{Y})$	$(X - \bar{X})(X - \bar{X})$
	1	70	-5	-210	1050	25
	3	140	-3	-140	420	9
	5	230	-1	-50	50	1
	8	350	2	70	140	4
	7	300	1	20	20	1
	12	570	6	290	1740	36
	8	380	2	100	200	4
	4	200	-2	-80	160	4
Mean:	6	280		Sum:	3780	84

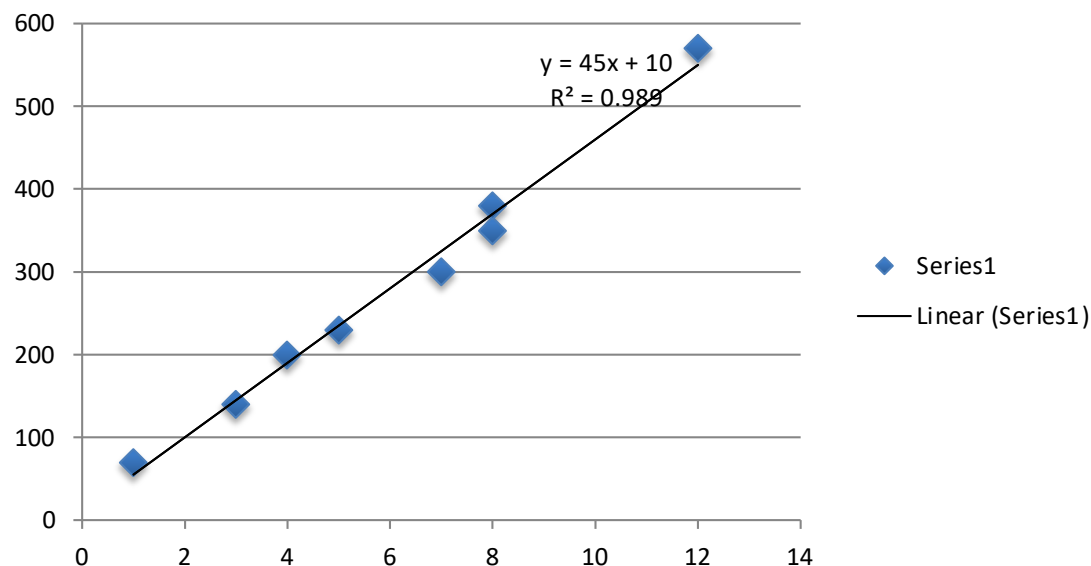
Example: least-squares



	X	Y	$X - \bar{X}$	$Y - \bar{Y}$	$(X - \bar{X})(Y - \bar{Y})$	$(X - \bar{X})(X - \bar{X})$	$\hat{y}$	Residual
	1	70	-5	-210	1050	25	55	15
	3	140	-3	-140	420	9	145	-5
	5	230	-1	-50	50	1	235	-5
	8	350	2	70	140	4	370	-20
	7	300	1	20	20	1	325	-25
	12	570	6	290	1740	36	550	20
	8	380	2	100	200	4	370	10
	4	200	-2	-80	160	4	190	10
Mean:	6	280		Sum:	3780	84		

Goodness of fit: R-squared

- How can we determine how well a regression line fits the data?
- To determine whether that line is a *good fit*, we can use the *coefficient of determination*
 - This is a measure denoted by R^2 (also written as *R-squared*)



R-squared: the coefficient of determination

- It is simply the square of the correlation coefficient (r) between the variable x and y and is defined as:

$$r^2 = \frac{(s_{xy})^2}{s_{xx} s_{yy}}$$

or

$$R^2 = r^2$$

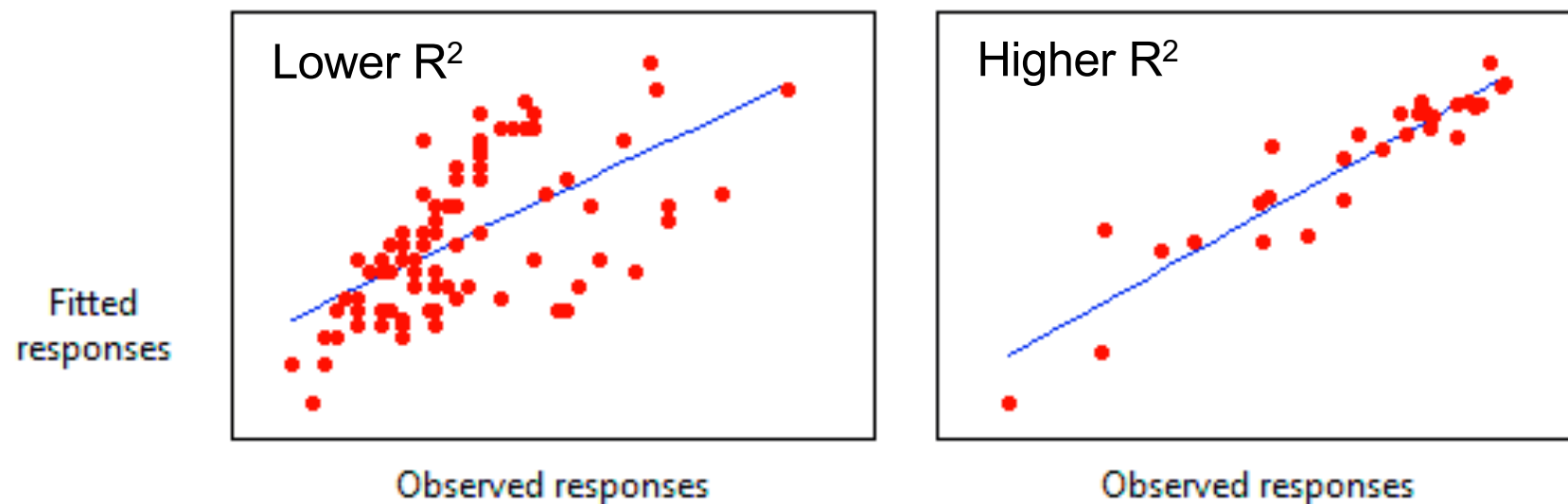
$$r = \frac{s_{xy}}{\sqrt{(s_{xx})(s_{yy})}}$$

It provides information as to how much of the variation in the y -variable (dependent variable) is attributable to (“caused by”) the x -variable (independent variable)

R-squared

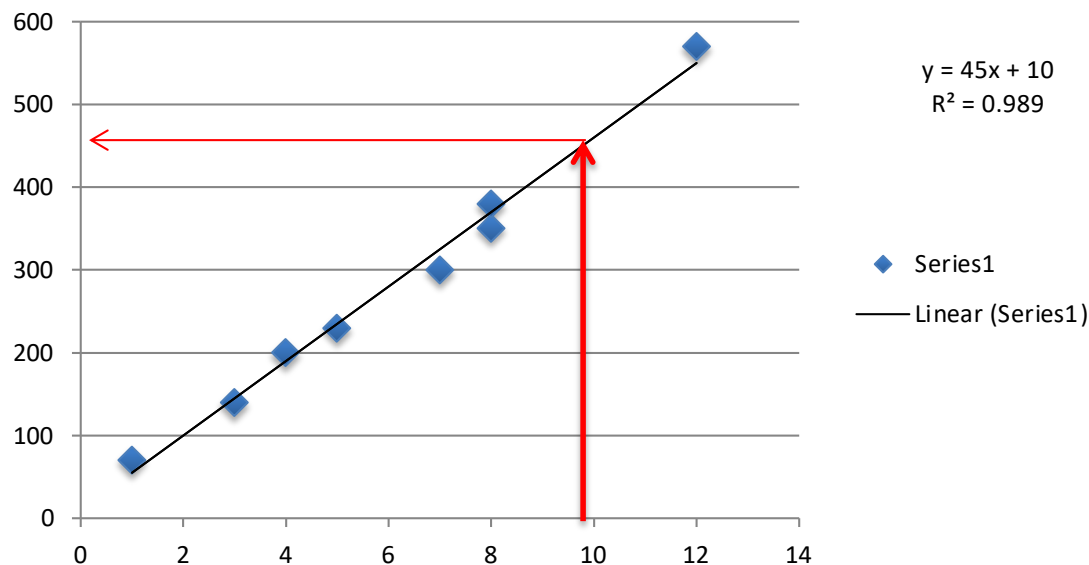
- Since the value of r always lies between -1 and 1 , the value of R^2 must always lie between 0 and 1
 - If the value of R^2 is close to 1 , a straight line fits the data well
 - If the value of R^2 is close to 0 , a straight line fits the data poorly

Plots of Observed Responses Versus Fitted Responses for Two Regression Models



Making Predictions: interpolation vs extrapolation

- Even if the regression line is a *good* fit of the data, it is dangerous to make a prediction of y for an x -value that is outside the range of the x -values used to determine the line equation
 - This is called extrapolation
- To make a simple ‘point’ prediction within range, substitute the x -value in question into the regression equation



$$\begin{aligned}x &= 10 \\ \hat{y} &= 10 + 45(10) \\ &= 460\end{aligned}$$

Example: Energy Use

For the temperature (celsius) and electricity consumption (kW) data the least squares regression line is:

$$\hat{y} = 63.2 - 1.30x$$

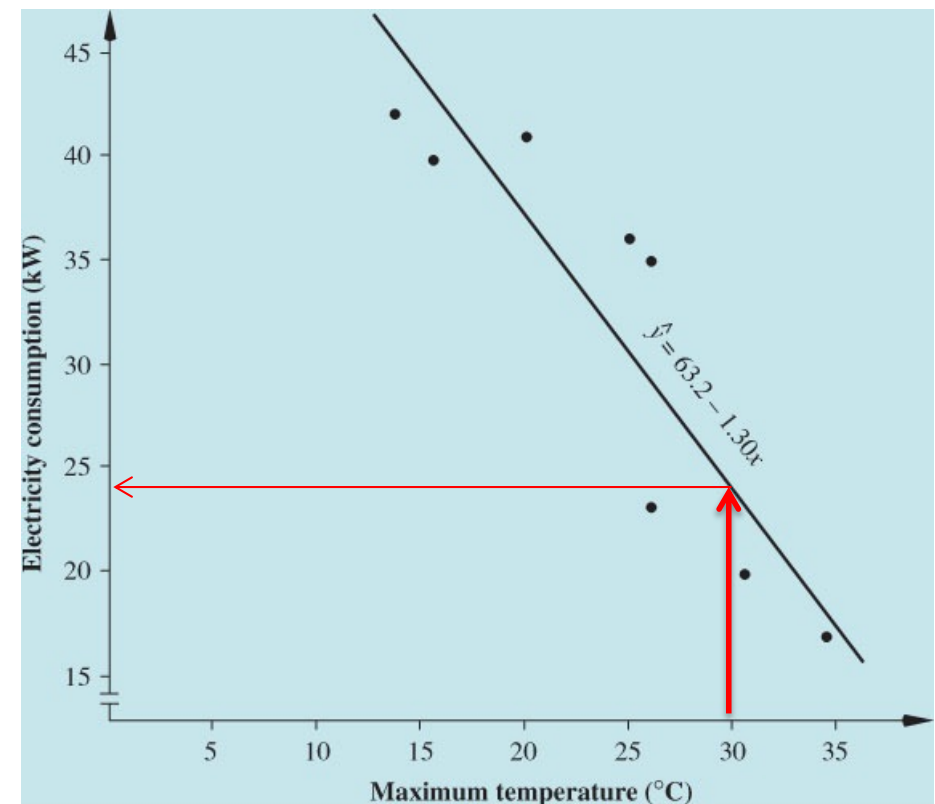
Predict the electricity consumption of office heating when the maximum temperature is 30°C

Solution:

Substituting $x = 30$

$$\begin{aligned}\hat{y} &= 63.2 - 1.30(30) \\ &= 24.2\end{aligned}$$

The regression line predicts that electricity consumption will be 24.2 kW @ 30°C

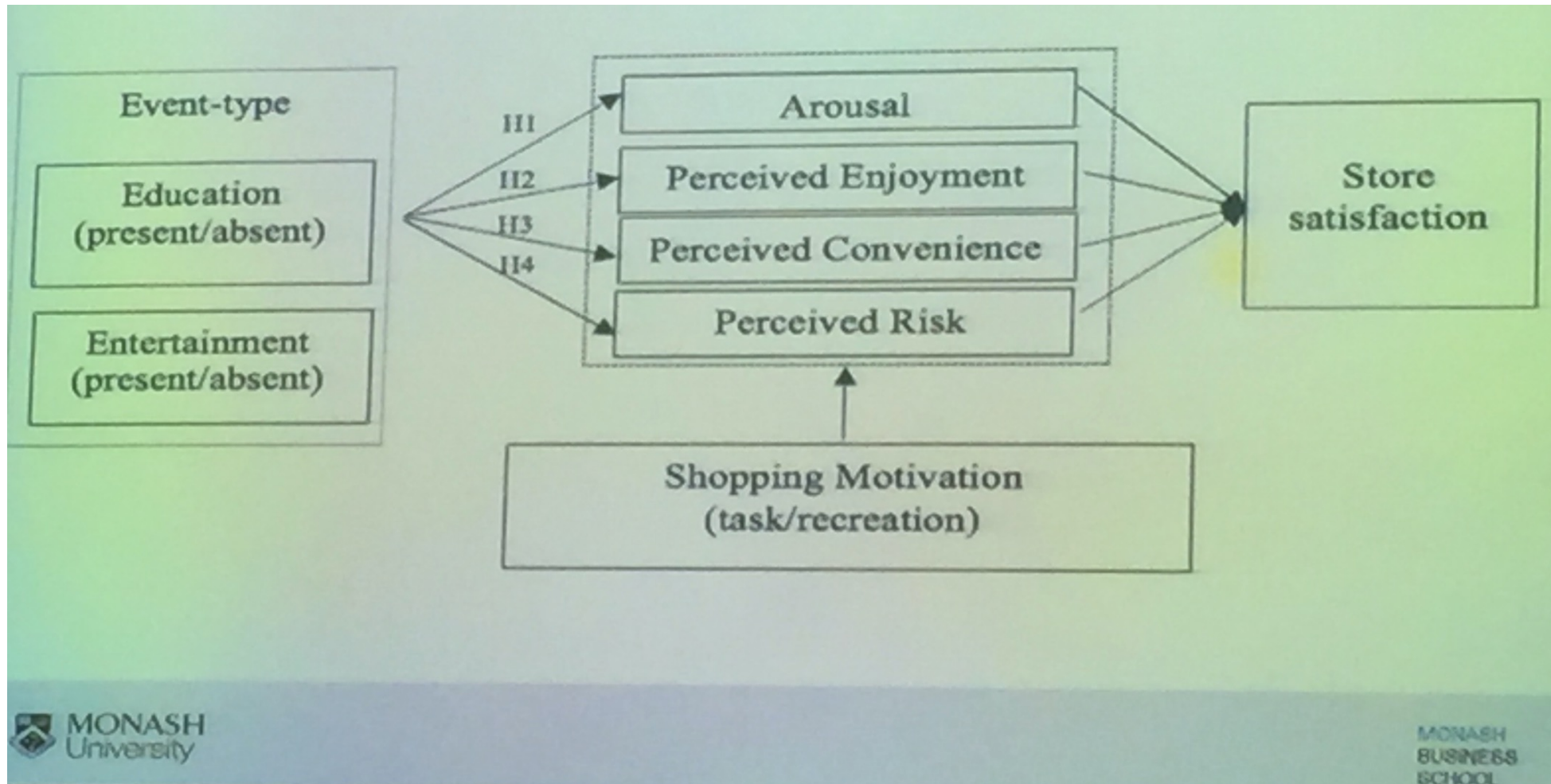


Multiple Regression

- The methods used in least-squares simple linear regression are appropriate only if a single *straight line* is the best fit
- For some data there may be other *non-linear* relationships (e.g. quadratic)
- In some cases, it may be desired to find the relationship between a dependent variable y and *several* independent x -variables, say $x_1, x_2, x_3, \dots, x_k$

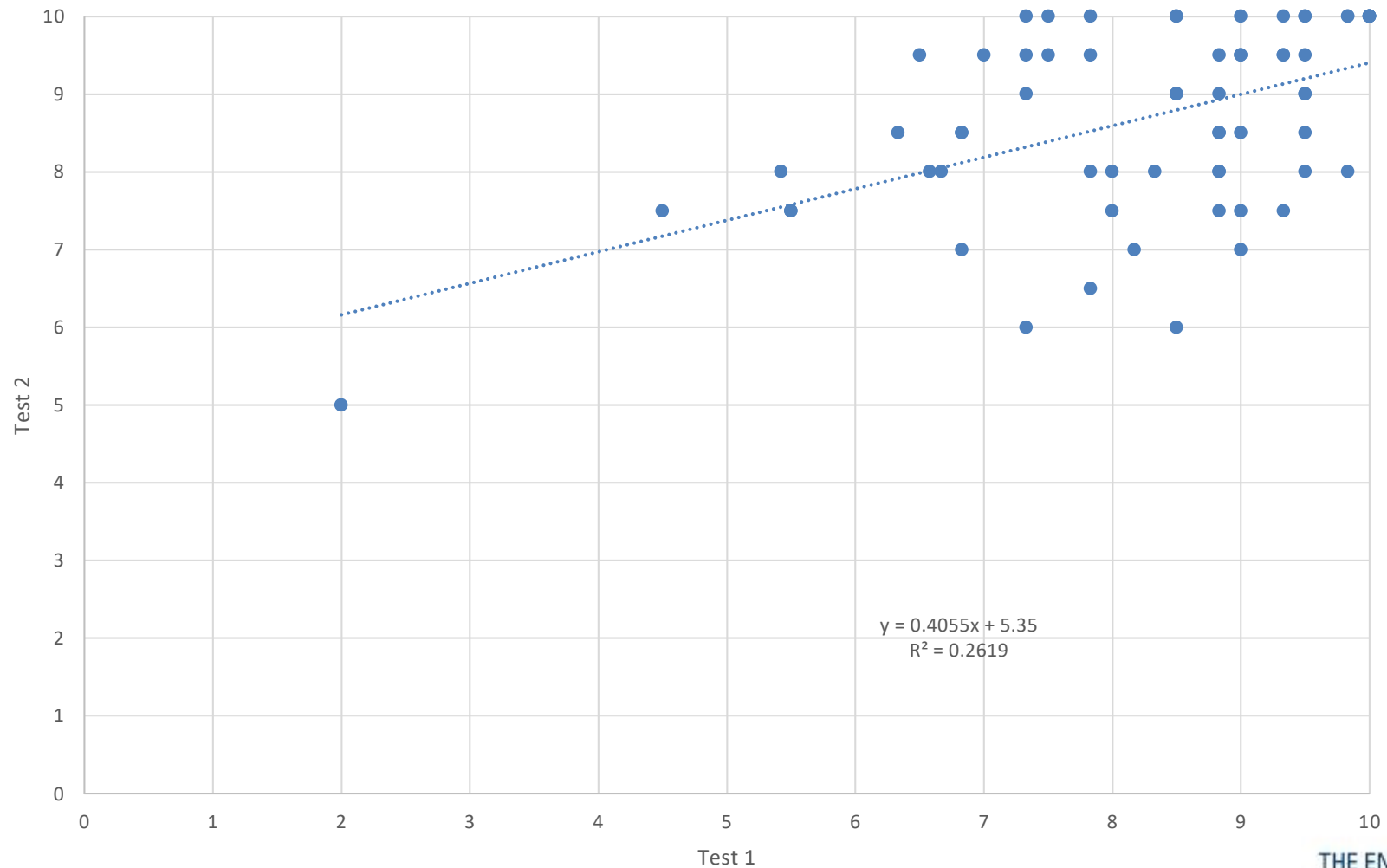
$$ROI_i = \frac{P(\text{resp})_i * P(q2r)_{i,sic} * P(\text{conv})_i * \text{Premium}_{i,sic} * \text{Commission}}{\text{Mailing cost}_i} \quad (2)$$

In-store Events → Store Satisfaction



Course Regression

Test 2 by Test 1



Single Regression: Excel Output

SUMMARY OUTPUT

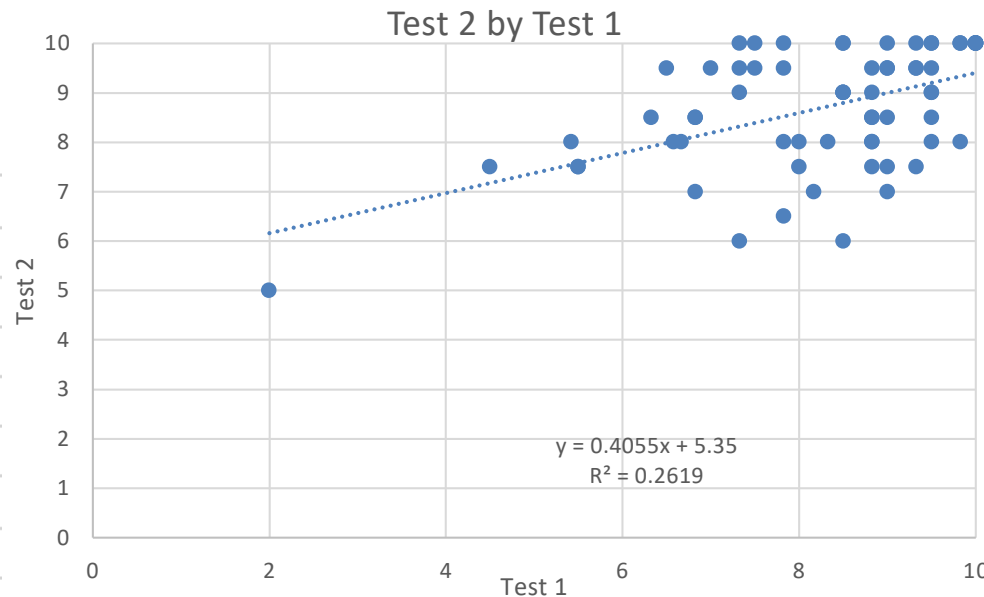
Regression Statistics

Multiple R	0.51176732
R Square	0.26190579
Adjusted R Square	0.25037306
Standard Error	1.03808023
Observations	66

ANOVA

	<i>df</i>	<i>SS</i>	<i>MS</i>	<i>F</i>	<i>Significance F</i>
Regression	1	24.472318	24.472318	22.7097978	1.1239E-05
Residual	64	68.9670759	1.07761056		
Total	65	93.4393939			

	<i>Coefficients</i>	<i>Standard Error</i>	<i>t Stat</i>	<i>P-value</i>	<i>Lower 95%</i>	<i>Upper 95%</i>	<i>Lower 95.0%</i>	<i>Upper 95.0%</i>
Intercept	5.35004698	0.71385568	7.49457784	2.5092E-10	3.92395631	6.77613764	3.92395631	6.77613764
X Variable 1	0.40550142	0.08509142	4.76547981	1.1239E-05	0.23551178	0.57549107	0.23551178	0.57549107



Multiple Regression: Excel Output

SUMMARY OUTPUT									
		Single Regression:							
Regression Statistics		Regression Statistics							
Multiple R	0.57456058	Multiple R	0.41529876						
R Square	0.33011986	R Square	0.17247306						
Adjusted R S	0.3086837	Adjusted R S	0.166008						
Standard Err	5.20304095	Standard Err	5.71478356						
Observations	130	Observations	130						
ANOVA									
	df	SS	MS	F	Significance F				
Regression	4	1667.62754	416.906884	15.4001368	2.8835E-10				
Residual	125	3383.95439	27.0716351						
Total	129	5051.58192							
	Coefficients	Standard Error	t Stat	P-value	Lower 95%	Upper 95%	Lower 95.0%	Upper 95.0%	
Intercept	9.90651777	2.51567199	3.93792108	0.00013574	4.92769068	14.8853449	4.92769068	14.8853449	
X Variable 1	0.82859919	0.29188556	2.838781	0.00528703	0.25092144	1.40627694	0.25092144	1.40627694	
X Variable 2	0.43957892	0.30149537	1.45799563	0.14734967	-0.1571178	1.03627567	-0.1571178	1.03627567	
X Variable 3	0.54394407	0.26815635	2.02845867	0.0446382	0.01322937	1.07465876	0.01322937	1.07465876	
X Variable 4	0.67996638	0.21087465	3.22450513	0.00161026	0.26261928	1.09731347	0.26261928	1.09731347	
X Variable 1	1.47857515	0.2862654	5.16505018	8.9619E-07	0.91215015	2.04500015	0.91215015	2.04500015	

Lab 9 Task 1

Task 1 step 1: For the following data, complete the manual calculation of the regression components, starting with the variable means

Data:	Test 1 mark	Test 2 mark	Complete Regression Components:				Predicted Y	Residual	Murphy, Andrew: Use the formula $\hat{y}=a+bX$, eg by using the cells below: =H\$29+H\$28*B7
	X	Y	$X - \bar{x}$	$Y - \bar{y}$	$(X - \bar{x})(Y - \bar{y})$	$(X - \bar{x})(X - \bar{x})$	$\hat{y}$	$y - \hat{y}$	
	9.8	8.0		8.0					
	8.5	10.0							
	8.8	9.0							
	7.5	10.0							
	5.4	8.0							
	4.0	8.0							
	9.5	9.0							
	6.8	7.0							
	6.8	8.5							
	7.5	9.5							
	8.0	7.5							
	9.0	10.0							
	8.8	8.0							
	8.5	9.0							
	2.0	5.0							
	4.5	7.5							
	8.0	8.0							
	9.3	7.5							
	10.0	10.0							
	8.5	9.0							
Mean:				Sum:	0	0		0.00E+00	← ≈ 0
	$(\bar{x})$	$(\bar{y})$		slope coefficient b:					
				intercept a:					
				Write complete Regression equation:					

Should match

Task 1 step 2: Now use Excel formulae to calculate the following:

	Estimate type	Excel formula	parameters hints
Correlation	Pearson coefficient (r)	=pearson()	array 1 = Ys; array 2 = Xs (separate with comma - can be other way around - does
Regression	slope coefficient (b)	=linest()	Ys, Xs order of Ys and Xs matters; 'const' and 'stats' are not needed here
	Intercept (a)	=intercept()	Ys, Xs order of Ys and Xs matters
	R squared (R^2)	=rsq()	Ys, Xs
	R squared (R^2)	=r * r	{alternative method}

Lab Task 1

Task 1 step 3: Construct a scatterplot of Y on X: Select cells B6:C26 and Insert >> Scatterplot

Hopefully the scatterplot will have Test 1 on the X (horizontal) axis and Test 2 on the Y (vertical)

Adjust both axes to 0-10 (double-click on an axis >> axis options >> axis bounds >> adjust minimum and maximum if necessary; repeat for other axis)

Then add the trendline: Chart Design >> Add Chart Element >> Trendline >> More Trendline Options >> Linear and

Paste chart below and interpret result.

☒ Display Equation on chart

☒ Display R-squared value on chart

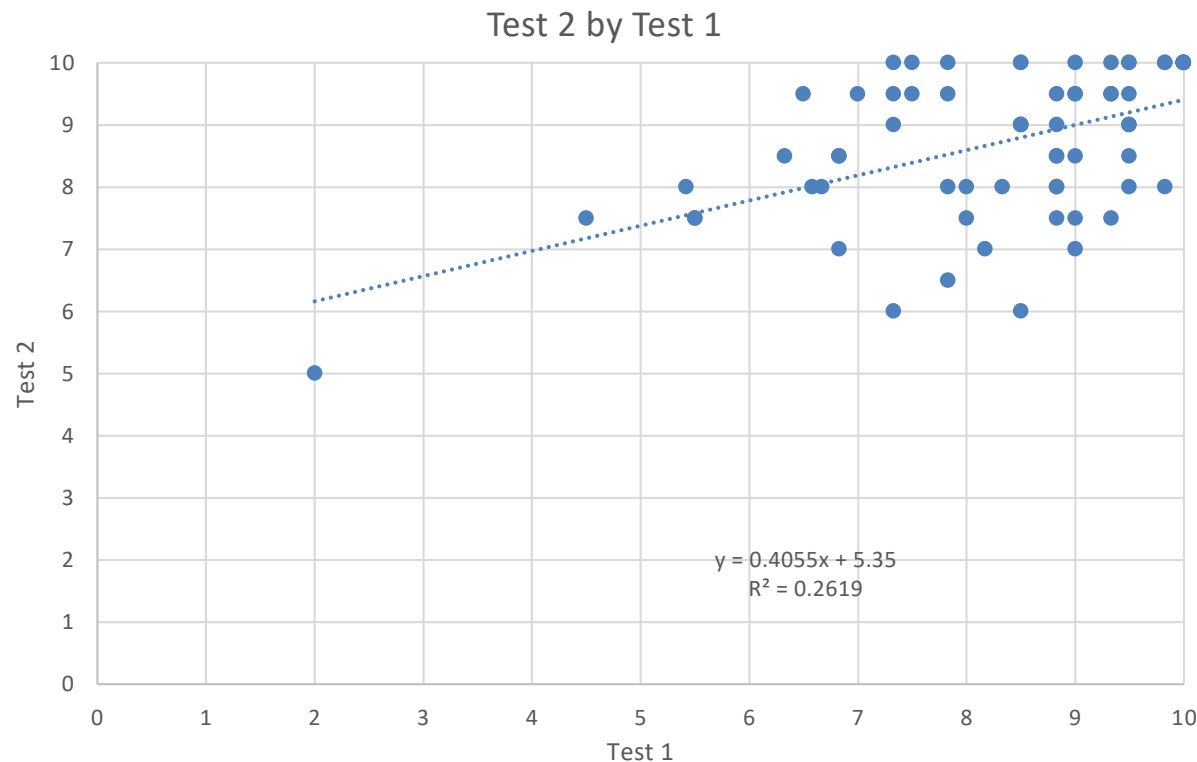
Murphy, Andrew:

If so, the series will look like:

'=SERIES('Lab 9'!\$C\$6,'Lab 9'!\$B\$7:\$B\$26,'Lab 9'!\$C\$7:\$C\$26,1).

Check it is right by looking for the first X value (\$B\$7) does it appear on the X axis?

If not, you need to click on the series, and edit the formula bar to reverse the order of \$B\$ and \$C\$



Lab Task 2

Murphy, Andrew:

Unfortunately you are unable to copy from the Remote App Browser version of Excel to a non-VM version, so there are two options:

- 1) use the Client version, which should allow you to copy or save;
- 2) open the Lab within the Virtual Machine version of Excel. This requires you to have your Lab saved in your H drive or Massey-provided OneDrive. You can do that by using the General Labs icon, and using Chrome to access Stream and the Lab, accessing Temple and transferring the data, and then upload the file to the Lab9_11 submission box as a temporary submission, and download it from there to your local machine.

The Client route is much easier, if you can install it

Create a new

Market:

Product:

Measure:

Time:

Task 2 Step 1: Use the IRI Temple plugin to extract data as per the IRI SKUs tab (see hints video in Topic 3)

Make sure the DistNum% variable is formatted with two decimal places. Paste the resulting table into the IRI SKUs tab of this file, and **remove the 10 products with 0 sales**.

Task 2 Step 2: Generate a Scatterplot with the remaining 69 SKUs (ignoring the Total Widgets row, if you have this), with **Dist Num %** on the X axis and **Dollars** on the Y

Add trendline and equation (with R-square statistic), **setting the Intercept to be 0** (since there can be no sales of Distribution% is 0)

Identify any very large Dollar values, and in a **second copy of the chart**, remove them. (Easiest way: sort descending by Sales\$, and exclude the top two rows of very large

Paste chart below

If you have trouble finding the data, having successfully loaded the plugin:
click on the plugin (IRI Temple) and then Data Source >> Open Drfx button >>
navigate to Data (E:) >> Temple Plugin >> database.drfx and Open.
Open the Database folder within and then Widgets

 Data Source Manager

✓ Devices and

Data (E

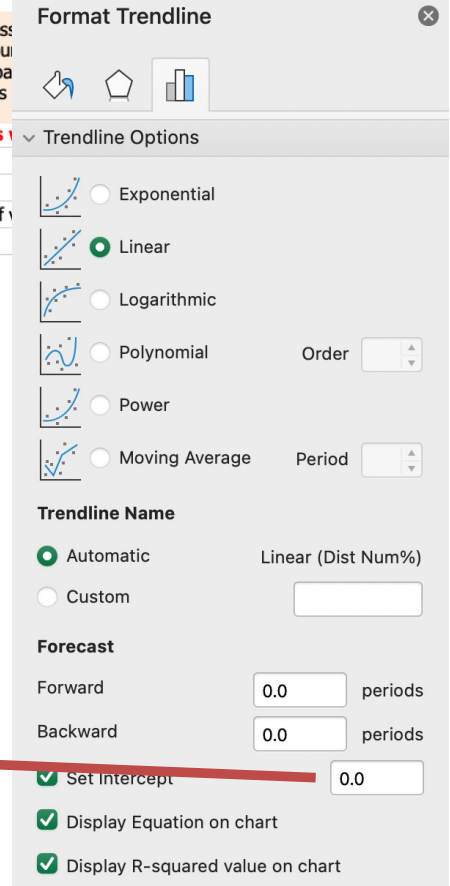
 917 MB[illegible]

(Moving columns around is required for Task 4)

	A	B	C	D	E	F	G	H	I	J
MAT To 19/04/26		New Zealand Grocery								
	Dollars	Units	Dist Num%	Price \$/Unit	Percentage Discount >= 3 % 9 wks	Highest Weekly Price	Lowest Weekly Price	Promoted Price >= 3 % 9 wks	Promoted Price \$/Unit >= 3 % 9 wks	Non Promoted Price \$/Unit
Better Rolls 1.2kg	15851018	2121056	99.02	7.47	10.5	8.76	4.39	6.83	8.35	
Better Rolls 750g	8338713	1473306	96.54	5.66	11.5	6.65	4.95	5.21	6.40	
Better Rolls Wild Neon 500g	6836847	953948	98.63	7.17	5.8	7.97	6.70	6.80	7.61	
Best Crunchy Pieces Super 500g	5888420	1457288	97.17	4.04	13.8	4.76	2.39	3.74	4.69	
Better Rolls 375g	3813094	810171	99.14	4.71	0.7	4.87	4.44	4.25	4.74	
Better Rolls 375g 1	2341834	309971	95.50	7.56	5.2	8.22	7.12	7.04	7.97	
Best Chippies 460g	2320096	515666	95.82	4.50	15.0	5.39	4.03	4.11	5.29	

ENGINE
THE NEW
V ZEALAND

Paste chart below



=SERIES('IRI SKUs'!\$D\$2,'IRI SKUs'!\$d\$3:\$d\$71,'IRI SKUs'!\$b\$3:\$b\$71,1)

Task 2 Step 1: Use the IRI Temple plugin to extract data as per the IRI SKUs tab (see hints video in Topic 3)

Make sure the DistNum% variable is formatted with two decimal places. Paste the resulting table into the IRI SKUs tab of this file, and **remove the 10 products**

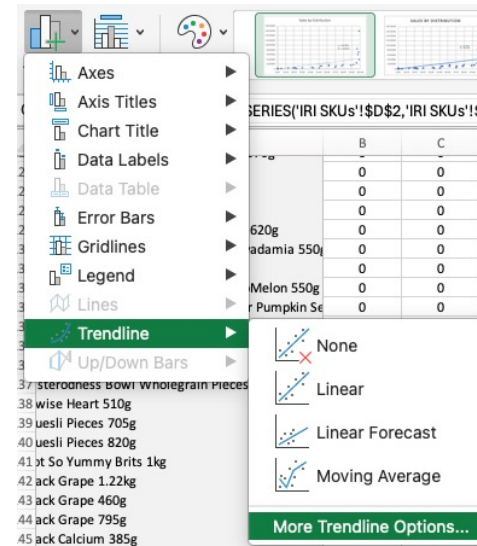
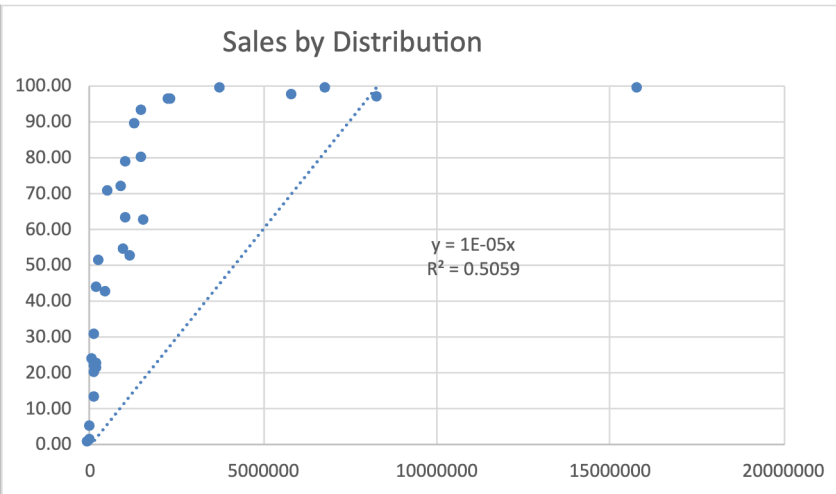
Task 2 Step 2: Generate a Scatterplot with the remaining 69 SKUs (ignoring the Total Widgets row, if you have this), with **Dist Num %** on the X axis and **Dollars** on the Y

Add trendline and equation (with R-square statistic), **setting the Intercept to be 0** (since there can be no sales of Distribution% is 0)

Identify any very large Dollar values, and in a **second copy of the chart**, remove them. (Easiest way: sort descending by Sales\$, and exclude the top two rows of

Paste chart below

If you have trouble finding the data, having success click on the plugin (IRI Temple) and then Data Source navigate to Data (E:) >> Temple Plugin >> database Open the Database folder within and then Widgets



Format Trendline

Trendline Options

- ☐ Exponential
- ☒ Linear
- ☐ Logarithmic
- ☐ Polynomial
- ☐ Power
- ☐ Moving Average

Trendline Name

☒ Automatic Linear (Dist Num%)

☐ Custom

Forecast

Forward: 0.0 periods

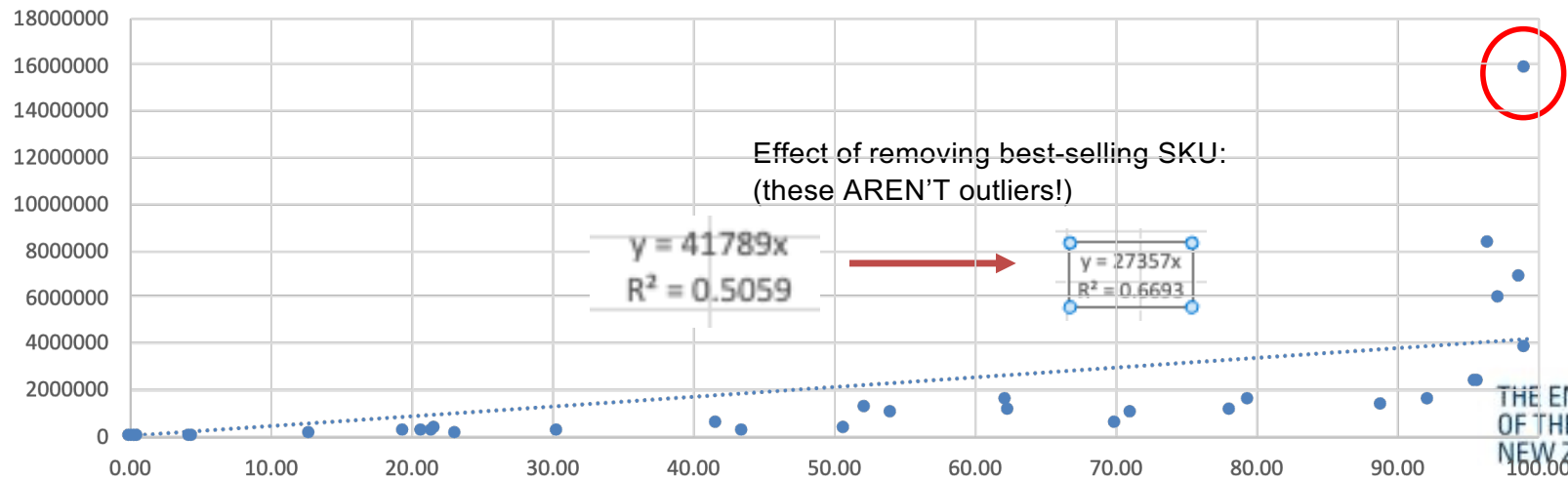
Backward: 0.0 periods

☒ Set Intercept 0.0

☒ Display Equation on chart

☒ Display R-squared value on chart

Sales by Distribution



Lab Task 3

Task 3: Use Data Analysis Toolpak function Regression to regress Dollars (DV) on Dist Num % (IV)

In regression speak, the "Y" variable (Dollars) is the "Dependent Variable" (DV)

The "X" variable (Dist Num %) is the "Independent Variable" (IV)

Run Regression in the IRI SKUs file, setting up as follows:

Copy and paste results below (or set B94 of this sheet as the output range)

Add in Analysis Toolpak: Mac: Tools :
PC: File > Options > Add-Ins > Manag
(This may need to be done each time
Once active, go to Data toolbar menu)

SUMMARY OUTPUT

Regression Statistics

Multiple R

0.000

R Square

0.000

Adjusted R Square

Standard Error

Observations

ANOVA

df

SS

MS

F

Significance F

Regression

Residual

Total

Coefficients

Standard Error

t Stat

P-value

Lower 95%

Upper 95%

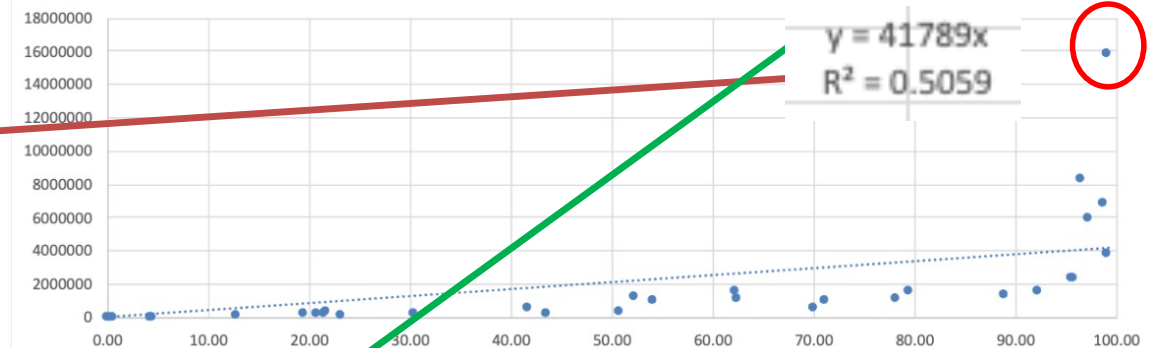
Lower 95.0% Upper 95.0%

Intercept

X Variable 1

DistNum%

Sales by Distribution



Lab 9 Task 4: Multiple Regression

		Task 3: Use Data Analysis Toolpak function Regression to regress Dollars (DV) on Dist Num % (IV)				PC: File > Options > Add-Ins > Manag (This may need to be done each time Once active, go to Data toolbar men											
		In regression speak, the "Y" variable (Dollars) is the "Dependent Variable" (DV) The "X" variable (Dist Num %) is the "Independent Variable" (IV) Run Regression in the IRI SKUs file, setting up as follows: Copy and paste results below (or set B94 of this sheet as the output range)															
		SUMMARY OUTPUT															
		Regression Statistics															
		Multiple R															
		0.000 R Square															
		0.000 Adjusted R Square															
		Standard Error															
		Observations															
		ANOVA															
		df		SS		MS		F		Significance F							
		Regression															
		Residual															
		Total															
		Coefficients		tandard Error		t Stat		P-value		Lower 95%		Upper 95%		Lower 95.0%		Upper 95.0%	
		Intercept															
		DistNum% X Variable 1															
		SUMMARY OUTPUT															
		Regression Statistics															
		Multiple R															
		0.402 R Square															
		0.339823463 Adjusted R Square															
		Standard Error															
		Observations															
		ANOVA															
		df		SS		MS		F		Significance F							
		Regression															
		Residual															
		Total															
		Coefficients		Standard Error		t Stat		P-value		Lower 95%		Upper 95%		Lower 95.0%		Upper 95.0%	
		Intercept															
		DistNum% (D)															
		Price \$/unit (E)															
		% Discount (F)															
		X Variable 1															
		X Variable 2															
		X Variable 3															

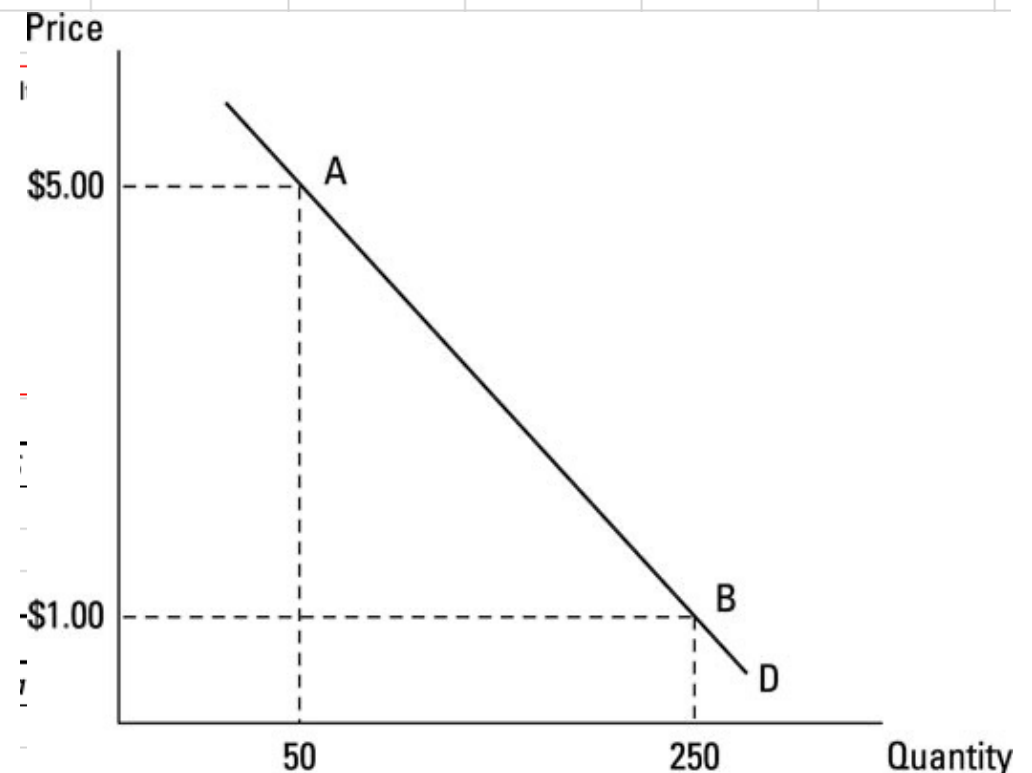
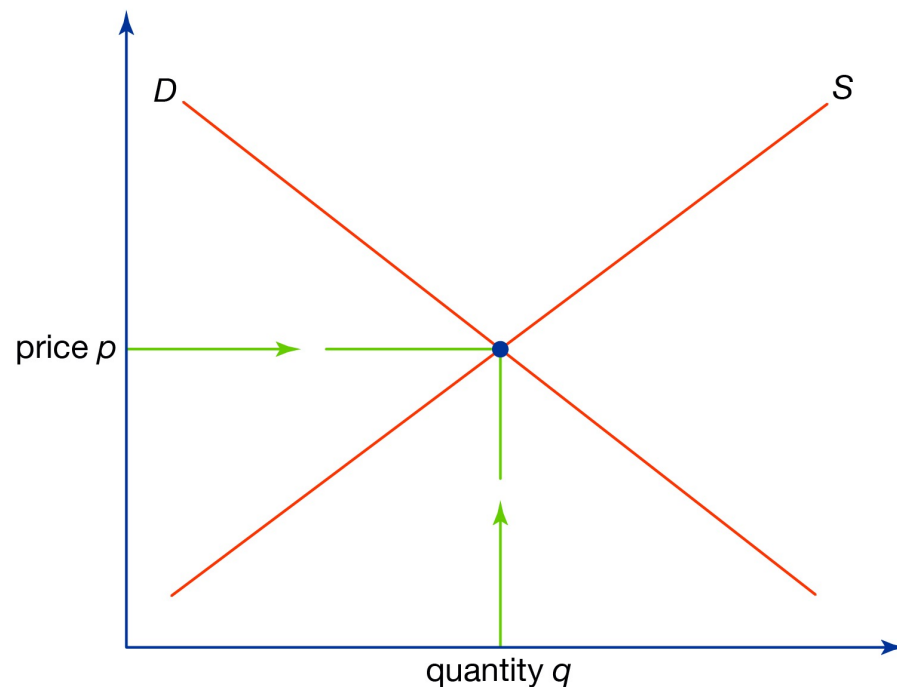
Is there “multicollinearity” between the explanatory variables (Xs)?

Lab 9 Task 5

Task 5: Use Data Analysis Toolpak function Regression to regress Units (DV) on Price \$/Unit (IV)

This is similar to the classic economics relationship: price and quantity, but done cross-sectionally: across a range of products
 In normal economics convention, price is put on the Y axis and quantity on the X (because Alfred Marshall did it that way in 1890)
 We will do it the regular ('marketing') way: regress Units (DV, Y, column C) on Price \$/unit (IV, X, column E)
 You do not need to rearrange the columns for this as there is only a single X. Again leave the constant blank
 Copy and paste results below (or set B149 in this sheet as the output range)

Supply and demand



Lab 9 Task 5

Task 5: Use Data Analysis Toolpak function Regression to regress Units (DV) on Price \$/Unit (IV)

This is similar to the classic economics relationship: price and quantity, but done cross-sectionally: across a range of products
In normal economics convention, price is put on the Y axis and quantity on the X (because Alfred Marshall did it that way in 1890)
We will do it the regular ('marketing') way: regress Units (DV, Y, column C) on Price \$/unit (IV, X, column E)
You do not need to rearrange the columns for this as there is only a single X. Again leave the constant blank
Copy and paste results below (or set B149 in this sheet as the output range)

179 **Task 5 step 2: create a Scatterplot of the relationship, including trendline and equations (use the raw data in the IRI SKUs tab)**

180 Ensure you have the Price per unit variable on the X-axis

Bonus: second chart excluding the top seller

